

1 Gain law

(bars: lengths by R ; $a_o := \Delta t / (\gamma_N R)$, $a_{\text{nom}} := a_o R$, $\bar{a} := a / a_{\text{nom}}$, $\bar{f} := a_o f$, $\bar{a} \bar{f} = \Delta \bar{w}$; $e_\theta = \theta - \hat{\theta}$; $\mathbb{E}[\cdot]$ over the noise; $P_{ij}[k] = \mathbb{E}[e_i[k] e_j[k]]$ posterior at k , P^f forecast; $\bar{w}_d[k]$ known)

$$\bar{a}_{\text{true}}(\bar{w}) = 1 - \frac{9}{8\bar{w}} + \frac{1}{2\bar{w}^3} - \frac{57}{100\bar{w}^4} + \frac{1}{5\bar{w}^5} + \frac{7}{200\bar{w}^{11}} - \frac{1}{25\bar{w}^{12}}$$

$$\bar{a}'_{\text{true}}(\bar{w}) = \frac{9}{8\bar{w}^2} - \frac{3}{2\bar{w}^4} + \frac{57}{25\bar{w}^5} - \frac{1}{\bar{w}^6} - \frac{77}{200\bar{w}^{12}} + \frac{12}{25\bar{w}^{13}}$$

$$\bar{a}''_{\text{true}}(\bar{w}) = -\frac{9}{4\bar{w}^3} + \frac{6}{\bar{w}^5} - \frac{57}{5\bar{w}^6} + \frac{6}{\bar{w}^7} + \frac{231}{50\bar{w}^{13}} - \frac{156}{25\bar{w}^{14}}$$

$$\begin{aligned} \bar{a}'[k] &:= \bar{a}'_{\text{true}}(\bar{w}_d[k] - \delta \hat{w}_3[k]), & \bar{a}''[k] &:= \bar{a}''_{\text{true}}(\bar{w}_d[k] - \delta \hat{w}_3[k]) \\ \frac{\partial \bar{a}'[k]}{\partial \bar{a}_w} &= 0, & \frac{\partial \bar{a}'[k]}{\partial \delta \bar{w}_3} &= -\bar{a}''[k], & a_w &= a_o \bar{a}_w \end{aligned}$$

2 Jacobians

(row 4 of F_e , first order in the step; the exact-map corrections, relative size $\bar{a}'[\Delta \bar{w}_d + (1 - \lambda_c) \delta \hat{w}_3] / (1 - \hat{a}_w) \approx 10^{-2}$, are not carried)

$$F_e(4, 3) = (1 - \lambda_c) \bar{a}'[k] - \bar{a}''[k] [\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k]]$$

$$F_e(4, 4) = 1 + F_{dw}[k] \bar{a}'[k]$$

3 True dynamics

$$(\delta \bar{w}_j[k] := \delta \bar{w}[k - d + j - 1], \quad j = 1, \dots, d + 1, \quad d = 2)$$

$$\bar{w}[k] = \bar{w}_d[k] - \delta \bar{w}_3[k]$$

$$\bar{w}_d[k + 1] = \bar{w}_d[k] + \Delta \bar{w}_d[k]$$

$$\delta \bar{w}_1[k + 1] = \delta \bar{w}_2[k]$$

$$\delta \bar{w}_2[k + 1] = \delta \bar{w}_3[k]$$

$$\delta \bar{w}_3[k + 1] = \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k]$$

$$\bar{w}[k + 1] - \bar{w}[k] = \Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k]$$

$$\bar{a}_w[k + 1] = \bar{a}_{\text{true}}(\bar{w}[k + 1])$$

$$F_{dw}[k] = \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k - i]$$

$$\varepsilon_w[k] = \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k - i] \bar{f}_T[k - i] + (1 - \lambda_c) n_w[k]$$

4 Process model

$$\mathbf{x} = [\delta\bar{w}_1, \delta\bar{w}_2, \delta\bar{w}_3, \bar{a}_w]^\top \quad (d = 2)$$

5 Estimator

$$\begin{aligned} \delta\hat{w}_1[k+1] &= \delta\hat{w}_2[k] \\ \delta\hat{w}_2[k+1] &= \delta\hat{w}_3[k] \\ \delta\hat{w}_3[k+1] &= \lambda_c \delta\hat{w}_3[k] \\ \hat{a}_w[k+1] &= \hat{a}_w[k] + \bar{a}'[k] [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]] \\ &\quad + \frac{1}{2} \bar{a}''[k] [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]]^2 \\ &\quad + \frac{1}{2} \bar{a}''[k] \left[(1 - \lambda_c)^2 P_{33}[k] + 2(1 - \lambda_c) F_{dw}[k] P_{34}[k] + F_{dw}[k]^2 P_{44}[k] + Q_{33}[k] \right] \\ &\quad - \bar{a}''[k] \left[(1 - \lambda_c) P_{33}[k] + F_{dw}[k] P_{34}[k] \right] \end{aligned}$$

($\ell_{31}[k]$: entry (3, 1) of $L[k]$, the y_1 gain of the update at k ; Measurement section. The n_w feedthrough adds no line: its four shares cancel, Error dynamics section.)

code: the first two terms of the $\hat{a}_w$ row are carried by the exact step,

$$\begin{aligned} &\frac{\bar{a}'[k](1 - \hat{a}_w[k]) [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]]}{(1 - \hat{a}_w[k]) + \bar{a}'[k] [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]]} \\ &= \bar{a}'[k] [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]] - \frac{\bar{a}'[k]^2}{1 - \hat{a}_w[k]} [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]]^2 + \mathcal{O}(3) \end{aligned}$$

whose second-order coefficient is $-\bar{a}'^2/(1 - \hat{a}_w)$ instead of $\frac{1}{2}\bar{a}''$; the difference $\frac{1}{2}(\bar{a}''[k] + 2\bar{a}'[k]^2/(1 - \hat{a}_w[k]))[\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]]^2$ is added together with the two mean lines (pred_mean2).

6 Error dynamics

(True – Estimator)

$$\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c e_{\delta\bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= [1 + F_{dw}[k] \bar{a}'[k]] e_{\bar{a}_w}[k] + \left[(1 - \lambda_c) \bar{a}'[k] - \bar{a}''[k] [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]] \right] e_{\delta\bar{w}_3}[k] \\
&\quad + \bar{a}'[k] \varepsilon_w[k] + (\text{second order, below})
\end{aligned}$$

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} \\ 0 & 0 & (1 - \lambda_c) \bar{a}' - \bar{a}'' [\Delta\bar{w}_d + (1 - \lambda_c) \delta\hat{w}_3] & 1 + F_{dw} \bar{a}' \end{bmatrix}, \quad \mathbf{q}[k] = [0, 0, -\varepsilon_w, \bar{a}'[k] \varepsilon_w]^\top$$

known and unknown parts of the true step, and the two starting points

$$\begin{aligned}
\bar{w}[k+1] - \bar{w}[k] &= [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]] + [(1 - \lambda_c) e_{\delta\bar{w}_3}[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k]] \\
\bar{w}[k] &= (\bar{w}_d[k] - \delta\hat{w}_3[k]) - e_{\delta\bar{w}_3}[k]
\end{aligned}$$

truth, second order about $\bar{w}_d[k] - \delta\hat{w}_3[k]$

$$\begin{aligned}
\bar{a}'_{\text{true}}(\bar{w}[k]) &= \bar{a}'[k] - \bar{a}''[k] e_{\delta\bar{w}_3}[k] + \mathcal{O}(e^2), \quad \bar{a}''_{\text{true}}(\bar{w}[k]) = \bar{a}''[k] + \mathcal{O}(e) \\
\bar{a}_w[k+1] - \bar{a}_w[k] &= \bar{a}'_{\text{true}}(\bar{w}[k]) [\bar{w}[k+1] - \bar{w}[k]] + \frac{1}{2} \bar{a}''_{\text{true}}(\bar{w}[k]) [\bar{w}[k+1] - \bar{w}[k]]^2 + \mathcal{O}(3) \\
&= \bar{a}'[k] [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]] + \bar{a}'[k] [(1 - \lambda_c) e_{\delta\bar{w}_3}[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k]] \\
&\quad - \bar{a}''[k] e_{\delta\bar{w}_3}[k] [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k] \\
&\quad + (1 - \lambda_c) e_{\delta\bar{w}_3}[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k]] \\
&\quad + \frac{1}{2} \bar{a}''[k] [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]]^2 \\
&\quad + \bar{a}''[k] [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]] [(1 - \lambda_c) e_{\delta\bar{w}_3}[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k]] \\
&\quad + \frac{1}{2} \bar{a}''[k] [(1 - \lambda_c) e_{\delta\bar{w}_3}[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k]]^2
\end{aligned}$$

estimator, second order (first two terms of the $\hat{a}_w$ row)

$$\hat{a}_w[k+1] - \hat{a}_w[k] = \bar{a}'[k] [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]] + \frac{1}{2} \bar{a}''[k] [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]]^2$$

truth – estimator

$$\begin{aligned}
e_{\bar{a}_w}[k+1] - e_{\bar{a}_w}[k] &= \bar{a}'[k] [(1 - \lambda_c) e_{\delta\bar{w}_3}[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k]] \\
&\quad - \bar{a}''[k] e_{\delta\bar{w}_3}[k] [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k] \\
&\quad + (1 - \lambda_c) e_{\delta\bar{w}_3}[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k]] \\
&\quad + \bar{a}''[k] [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]] [(1 - \lambda_c) e_{\delta\bar{w}_3}[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k]] \\
&\quad + \frac{1}{2} \bar{a}''[k] [(1 - \lambda_c) e_{\delta\bar{w}_3}[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k]]^2 + \mathcal{O}(3)
\end{aligned}$$

mean, with A1 $\mathbb{E}[e_{\delta\bar{w}_3}] = \mathbb{E}[e_{\bar{a}_w}] = 0$, A2 $\mathbb{E}[\delta\hat{w}_3 e_{\delta\bar{w}_3}] = \mathbb{E}[\delta\hat{w}_3 e_{\bar{a}_w}] = 0$, A3 $\mathbb{E}[\varepsilon_w e_{\bar{a}_w}] = 0$, $\mathbb{E}[\varepsilon_w^2] = Q_{33}$, and (the update at k and the controller read the same $y_1[k]$: $\delta\hat{w}_3[k] \ni \ell_{31}[k] n_w[k]$, $e_{\delta\bar{w}_3}[k] \ni -\ell_{31}[k] n_w[k]$, $\varepsilon_w[k] \ni (1 - \lambda_c) n_w[k]$, $\bar{a}'[k] = \bar{a}'_{\text{true}}(\bar{w}_d[k] - \delta\hat{w}_3[k]) \ni -\bar{a}''[k] \ell_{31}[k] n_w[k]$)

$$\begin{aligned}\mathbb{E}[\delta\hat{w}_3[k] \varepsilon_w[k]] &= (1 - \lambda_c) \ell_{31}[k] R_1 \\ \mathbb{E}[e_{\delta\bar{w}_3}[k] \varepsilon_w[k]] &= -(1 - \lambda_c) \ell_{31}[k] R_1 \\ \mathbb{E}[\bar{a}'[k] \varepsilon_w[k]] &= -\bar{a}''[k] (1 - \lambda_c) \ell_{31}[k] R_1\end{aligned}$$

$$\begin{aligned}\mathbb{E}[e_{\bar{a}_w}[k+1]] - \mathbb{E}[e_{\bar{a}_w}[k]] &= -\bar{a}''[k] (1 - \lambda_c) \ell_{31}[k] R_1 \\ &\quad - \bar{a}''[k] \left[(1 - \lambda_c) P_{33}[k] + F_{dw}[k] P_{34}[k] \right] + \bar{a}''[k] (1 - \lambda_c) \ell_{31}[k] R_1 \\ &\quad + \bar{a}''[k] (1 - \lambda_c)^2 \ell_{31}[k] R_1 \\ &\quad + \frac{1}{2} \bar{a}''[k] \left[(1 - \lambda_c)^2 P_{33}[k] + 2(1 - \lambda_c) F_{dw}[k] P_{34}[k] + F_{dw}[k]^2 P_{44}[k] + Q_{33}[k] \right] \\ &\quad - \bar{a}''[k] (1 - \lambda_c)^2 \ell_{31}[k] R_1 \\ &= -\bar{a}''[k] \left[(1 - \lambda_c) P_{33}[k] + F_{dw}[k] P_{34}[k] \right] \\ &\quad + \frac{1}{2} \bar{a}''[k] \left[(1 - \lambda_c)^2 P_{33}[k] + 2(1 - \lambda_c) F_{dw}[k] P_{34}[k] + F_{dw}[k]^2 P_{44}[k] + Q_{33}[k] \right]\end{aligned}$$

(lines 1–4 in the order of the truth – estimator expansion; the four $\ell_{31}R_1$ shares sum to zero)

compensation: the two lines added to $\hat{a}_w[k+1]$ (Estimator section)

$$\mathbb{E}[e_{\bar{a}_w}[k+1]] - \mathbb{E}[e_{\bar{a}_w}[k]] = 0$$

code: with the exact step in place of $\bar{a}'[\cdot] + \frac{1}{2}\bar{a}''[\cdot]^2$ the mean gains $\frac{1}{2}(\bar{a}''[k] + 2\bar{a}'[k]^2/(1 - \hat{a}_w[k]))(\Delta\bar{w}_d[k] + (1 - \lambda_c)\delta\hat{w}_3[k])^2$, which `pred_mean2` adds back.

7 Measurement and innovation

$$\begin{aligned}\nabla_d \bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k-d] \\ y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\ y_2[k] &= \bar{a}_w[k-d] + n_a[k], \quad \text{Var}(n_a) = R_2 \\ \bar{a}_w[k-d] &= \bar{a}_w[k] - \bar{a}'[k] \nabla_d \bar{w}_d[k] \quad (\text{first order, as in the code})\end{aligned}$$

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad L[k] = P^f H^\top (H P^f H^\top + R)^{-1}$$

$$e_{y_1}[k] = e_{\delta\bar{w}_1}[k] + n_w[k], \quad e_{y_2}[k] = e_{\bar{a}_w}[k] + n_a[k]$$

8 Process and measurement noise

$$\begin{aligned}\kappa_T &:= \frac{4k_B T a_o}{R}, & \text{Var}(\bar{a}_w[k] \bar{f}_T[k]) &= \kappa_T \bar{a}_w[k] \rightarrow \kappa_T \hat{\bar{a}}_w[k], & \sigma_{n_w}^2 &= \frac{\sigma_n^2}{R^2} \\ Q_{33} &= \kappa_T \left(\hat{\bar{a}}_w[k] + (1 - \lambda_c)^2 (\hat{\bar{a}}_w[k-1] + \hat{\bar{a}}_w[k-2]) \right) + (1 - \lambda_c)^2 \sigma_{n_w}^2 \\ Q_{34} &= Q_{43} = -\bar{a}'[k] Q_{33}, & Q_{44} &= \bar{a}'[k]^2 Q_{33}\end{aligned}$$

$$\begin{aligned}R &= \text{diag}(R_1, R_2), & R_1 &= \sigma_{n_w}^2, & R_2 &= 2a_{\text{cov}}^2 IF_{\text{var}}(\hat{\bar{a}}_w) (\hat{\bar{a}}_w + \xi)^2 \\ IF_{\text{var}}(\hat{\bar{a}}_w) &= 1 + 2 \left[c_1 u^2 + 2c_2 u(1-u) + c_3(1-u)^2 \right], & u &= \frac{\hat{\bar{a}}_w}{\hat{\bar{a}}_w + \xi}, & \xi &= \rho_n \frac{\sigma_{n_w}^2}{\kappa_T}\end{aligned}$$

$$(c_1 = 1.42044, c_2 = 0.0098975, c_3 = 0.098910, \rho_n = 0.350941, a_{\text{cov}} = 0.05)$$

Hold level mode

($\Delta \bar{w}_d = 0$, $\nabla_d \bar{w}_d = 0$; $\mu_3 := \mathbb{E}[e_{\delta \bar{w}_3}]$, $\mu_4 := \mathbb{E}[e_{\bar{a}_w}]$, $\iota_1 := \mathbb{E}[e_{y_1}]$, $\iota_2 := \mathbb{E}[e_{y_2}]$, $\bar{F} := \mathbb{E}[F_{dw}]$, $g := \text{Cov}(F_{dw}, e_{\bar{a}_w}) + \mathbb{E}[\varepsilon_w]$, $\beta_2 := \text{readout level offset}$, $r := \text{predict mean residual per step}$, $\gamma := a_{\text{cov}}(1-S)$; superscript $-$ = before the update)

$$\begin{aligned}\mu_3^-[k+1] &= \lambda_c \mu_3[k] - \bar{F} \mu_4[k] - g \\ \mu_4^-[k+1] &= \mu_4[k] + \bar{a}'(1 - \lambda_c) \mu_3[k] + \bar{a}' \bar{F} \mu_4[k] + \bar{a}' g + r \\ \iota_1[k+1] &= \mu_3^-[k+1] \\ \iota_2[k+1] &= a_{\text{cov}}[(1-S) \mu_4^-[k+1] + \beta_2] \\ \mu_3[k+1] &= \mu_3^-[k+1] - \ell_{31} \iota_1[k+1] - \ell_{32} \iota_2[k+1] \\ \mu_4[k+1] &= \mu_4^-[k+1] - \ell_{41} \iota_1[k+1] - \ell_{42} \iota_2[k+1]\end{aligned}$$

steady state ($\ell_{32} = 0$)

$$\begin{aligned}D &= 1 - (1 - \ell_{31}) \lambda_c, & E &= \ell_{41} + \bar{a}' \ell_{31} (1 - \ell_{42} \gamma) \\ \mu_3^- &= -\frac{\bar{F} \mu_4 + g}{D}, & \mu_3 &= (1 - \ell_{31}) \mu_3^- \\ \kappa \mu_4 &= s, & \kappa &= \ell_{42} \gamma - \frac{E \bar{F}}{D}, & s &= \frac{E g}{D} + (1 - \ell_{42} \gamma) r - \ell_{42} a_{\text{cov}} \beta_2\end{aligned}$$

$$\begin{aligned}\bar{a}'(1 - \lambda_c) \mu_3 + \bar{a}' \bar{F} \mu_4 + \bar{a}' g &= -\bar{a}'(1 - \lambda_c) \mathbb{E}[\delta \hat{w}_3] & (\text{stationary hold: } \mathbb{E}[\bar{a}_w[k+1] - \bar{a}_w[k]] = 0) \\ \text{two-step delay kept: } D &\rightarrow (1 + \ell_{21})(1 - \lambda_c) + \ell_{31}\end{aligned}$$

$$g = r = \bar{F} = 0 \Rightarrow \mu_4 = -\frac{\beta_2}{1 - S}$$